

SaVaGe User Manual

Product: SaVaGe — image-to-SVG converter and vector editor for Windows
Document version: matches application v0.1.0

How to use this manual

If you are...	Start here
New to vector graphics	Quick start → Convert → Edit basics
Comfortable with Illustrator / Inkscape	Workspace map → Tools reference → Advanced
Looking up a menu or key	Menus → Keyboard shortcuts

Inside SaVaGe, **Help** → **User Manual (PDF)...** opens this guide in your usual PDF viewer.

Keyboard shortcuts appear in **bold** (for example **V** for Select). Menu paths use **File** → **Open...** style.

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1. Quick start

1.1 Launch

Open SaVaGe the same way you open any Windows app:

- Double-click the **SaVaGe** shortcut on the desktop or in the Start menu, or
- Double-click **SaVaGe.exe** in the folder where you installed it.

SaVaGe opens in **Convert** mode: dark graphite window, lime accent, SaVaGe wordmark.

You do not need a terminal, PowerShell, or developer tools to run the program.

1.2 Your first conversion

1. Drag a PNG, JPEG, WEBP, GIF, BMP, or TIFF onto the drop zone — or click **Open Image....**
2. Pick a preset that matches the art:
 - **Logo / flat** — logos, icons, flat color art
 - **Photo** — photographs and soft gradients
 - **Line art** — sketches, comics, high-contrast line drawings
 - **Pixel** — pixel art / hard block shapes
3. Optionally tweak **Color precision**, **Filter speckle**, **Corner threshold**, **Path precision**, and **Mode**.
4. Click **Convert to SVG**.
5. Check the side-by-side **Raster | SVG** preview.
6. Click **Open in Editor** (or switch the toolbar to **Edit**) to refine the vectors.

You can also start from **File** → **Open....**: choosing a photo or PNG switches you to Convert and attaches that file automatically.

1.3 Your first edit

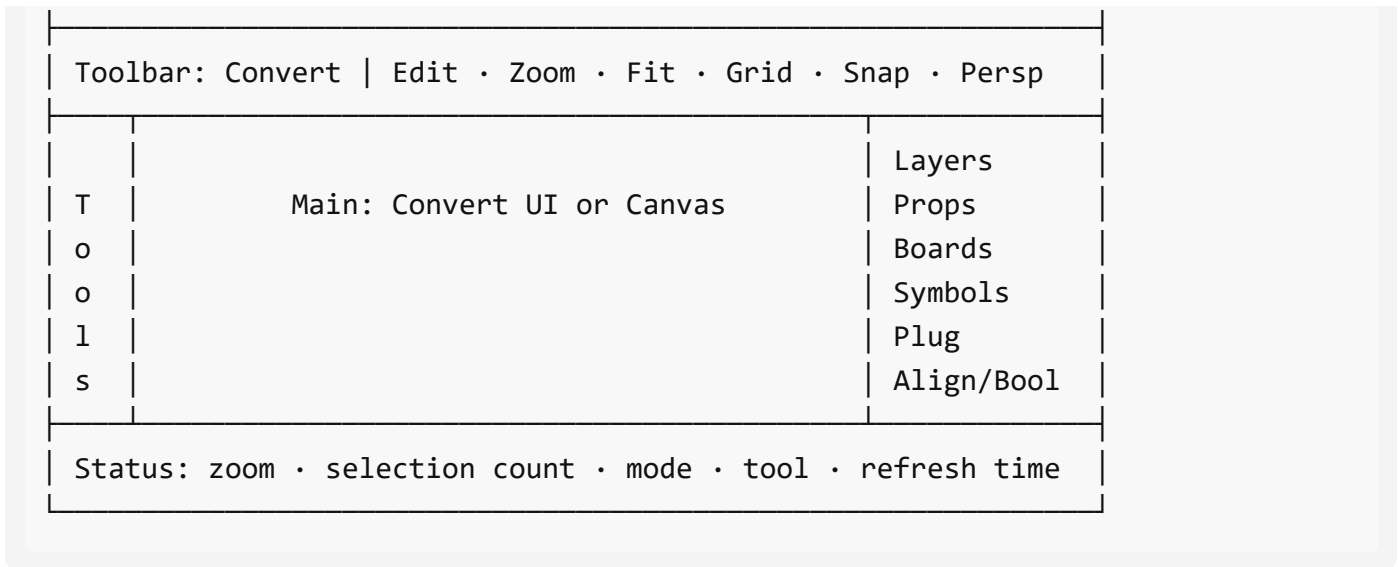
1. Press **V** (Select). Click a shape; drag to move. Use corner and edge handles to resize; use the rotate handle to rotate.
2. Open the **Props** tab on the right. Change fill color or opacity. **W** and **H** set the object's on-canvas size.
3. **File** → **Save Project...** and save a **.savage** file (best fidelity).
Or **File** → **Export SVG...** / **Export PNG...** for delivery.

1.4 Five-minute power path

Convert with **Logo** preset → **Open in Editor** → Unite overlapping paths → Shape Builder (**S**) to carve → Create Symbol → Mesh fill on a hero shape → Export SVG and keep a **.savage** master.

2. Workspace map

Title bar: SaVaGe · File Edit Object View Help



The left tool rail is active in **Edit** mode only.

2.1 Modes

Mode	Purpose
Convert	Import a bitmap and trace it to editable SVG
Edit	Draw, style, boolean, symbols, and export

Switch with the **Convert | Edit** control in the toolbar. After a successful trace, stay on Convert to compare the preview, then click **Open in Editor** when you are ready.

2.2 Status bar

The bottom strip shows:

- Zoom percentage
- How many objects are selected
- Current mode and active tool
- Last screen-refresh time (turns amber when the drawing is heavy)

Short confirmation messages (boolean results, clip masks, plugin actions) appear as toasts over the main view.

3. Convert mode

Convert turns bitmaps into editable vector paths.

3.1 Loading an image

Method	Notes
Drag and drop	Drop a raster file onto the large drop zone
Open Image...	File dialog for png / jpg / jpeg / gif / webp / bmp / tif / tiff
File → Open...	Same raster types; SaVaGe switches to Convert and attaches the file

A thumbnail of the source appears after the file is attached. While converting, the drop zone is disabled and a progress indicator appears. Non-image files dropped on Convert are ignored.

3.2 Presets (start here)

Presets apply a full option set. Choose the closest match, then fine-tune.

Preset	Best for	Tendency
Logo / flat	Logos, icons, UI graphics	Solid colors, light cleanup, smooth curves
Photo	Photos, soft shading	Fewer colors, stronger speck cleanup
Line art	Ink drawings, comics	High-contrast, few colors
Pixel	Pixel art	Blocky 1:1 pixel shapes

3.3 Trace options

Control	What it does	Tip	Power user
Color precision	How many distinct colors to keep	Higher = more colors and a heavier file	Lower for flat brand marks; raise for photos
Filter speckle	Removes tiny islands of noise	Increase if the result looks speckled	Clean up here before Simplify Path
Corner threshold	Corner vs curve bias	Default is fine for logos	Lower for sharper corners; higher for smoother bends
Path precision	How tightly paths follow the original	Leave default unless the SVG is huge	Lower to reduce point count
Mode	Spline / Polygon / Pixel	Use Spline for logos	Polygon for a faceted look; Pixel for 1:1 blocks

A few extra engine settings come with each preset and are not shown as separate sliders.

3.4 Preview and continue

After conversion you get a side-by-side **Raster | SVG** preview. The traced drawing is already loaded into the document.

- Click **Open in Editor** to refine it on the canvas.
- Or switch the toolbar to **Edit** at any time.
- Stay on Convert if you want to change options and convert again before editing.

3.5 Convert workflow tips

- Start with the smallest useful image that still has clean edges (very large photos produce heavy path sets).
 - Logo art with solid fills converts cleaner than soft, anti-aliased shadows.
 - After convert, use **Object** → **Simplify Path** on noisy selections before booleans.
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4. Edit mode basics

4.1 Navigating the canvas

Action	How
Pan	H (Pan tool), hold Space + drag, or middle-mouse drag
Zoom toward cursor	Scroll wheel
Zoom tool	Z — click to zoom in, Alt-click to zoom out
Zoom buttons	Toolbar – / % / +
Fit active artboard	Toolbar Fit, View → Fit Artboard , or Ctrl+0
Fit selection	View → Fit Selection or Ctrl+2
Zoom 100%	Click the zoom %, View → Zoom 100% , or Ctrl+1
Zoom 50% / 200%	View menu

4.2 Selecting and transforming

1. Activate **Select (V)**.
2. Click an object. **Shift+click** to add or remove from the selection.
3. Drag the object to move it.
4. Drag handles to resize. The opposite edge or corner stays put. **Shift** keeps proportions.
5. Use the rotate handle to rotate around the selection's center.
6. Arrow keys nudge 1 px; **Shift+arrows** nudge 10 px.

Ctrl+A selects all. **Delete** or **Backspace** deletes the selection. While you are typing in a field (Properties, text overlay, and similar), those keys edit the field instead of the canvas.

Clicking empty canvas clears the selection. Locked objects cannot be clicked; unlock them in Layers first.

4.3 Undo and history

Action	Shortcut / menu
Undo	Ctrl+Z · Edit → Undo
Redo	Ctrl+Shift+Z or Ctrl+Y · Edit → Redo

A full drag, stroke, or resize counts as one undo step. Switching tools while you are still dragging finishes that shape first.

4.4 Layers at a glance

Open the **Layers** tab:

- Top of the list $\approx$ front of the stack
- Click to select
- Toggle visibility and lock
- Double-click to rename
- Drag to reorder among top-level items
- Groups show nested children indented

4.5 Properties at a glance

Open **Props** (it always inspects the **first** object in the selection):

- Position **X / Y**, size **W / H**, rotation **R**, opacity
- Fill and stroke
- Stroke width
- Effects: blur, drop shadow
- Extra controls for text and symbols when those are selected

Changing **W** or **H** scales the object to that size on the canvas.

5. Tools reference

Tools live in the left rail (Edit mode only). Click an icon or use a shortcut. Hover an icon to see its name.

A click without a drag is ignored for drawing tools (no leftover 1-pixel specks). Switching to another tool while drawing finishes the current shape.

5.1 Navigation & selection

Select — V

Click objects to select them. Move, multi-select (**Shift**), scale, and rotate with the handles around the selection.

Direct Select — A

Edits individual path points. Click near an anchor (on any path) to select that path and drag the point. Curve handles travel with the point.

Variable width: hold **Alt** and drag a point to change that point's local stroke width. See [Variable-width strokes](#).

Pan — H

Drag the canvas. Hold **Space** for a temporary pan without leaving your current drawing tool.

Zoom — Z

Click the canvas to zoom in around the pointer. **Alt-click** zooms out. The scroll wheel still zooms toward the cursor at any time.

5.2 Shape tools

Tool	Shortcut	Draw	Modifiers
Rectangle	R	Drag	Shift square · Alt from center
Ellipse	O	Drag	Shift circle · Alt from center
Line	L	Drag	Shift snaps to horizontal, vertical, or 45°
Polygon	—	Drag out the radius	Regular hexagon (6 sides)
Star	—	Drag out the radius	5-point star

New shapes use the lime accent fill where it fits the tool.

5.3 Path tools

Pen — P

Click to place corner anchors. Drag while placing to create Bézier handles. Click near the first point (the snap distance stays the same at any zoom) to close the path.

Key	Action
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Enter or Esc	Finish the path (leave it on the canvas)
Backspace	Remove the last point (does not delete the whole path)

Leaving the Pen tool also finishes the current path so the next Pen session starts fresh.

Pencil — N

Freehand stroke, simplified into an open path. Good for roughing silhouettes; refine with Direct Select or **Simplify Path**. Grid snap is off while you draw so the stroke follows your hand.

Brush (calligraphy) — B

Speed-sensitive calligraphy: faster motion → thinner ribbon. The result is a **closed filled path** (not a simple outline), so fills and booleans treat it as solid geometry. Grid snap is off while you draw.

5.4 Pattern Brush & Scatter Brush

Tool	Result
Pattern Brush	Chevron motifs stamped along your stroke at regular spacing, grouped
Scatter Brush	Leaf-like motifs with position jitter, rotation, and scale variation, grouped

How to use: choose the tool → drag a gesture → release. SaVaGe creates a group of stamp paths. A click without a drag does nothing. Grid snap is off while you stroke.

Stamps are ordinary vector paths. Ungroup if you need to edit or boolean individual stamps. Spacing and jitter are fixed in this release (not yet shown as sliders).

5.5 Shape Builder — S

Interactive region sculpting for overlapping filled shapes (rectangles, ellipses, closed paths).

1. Select two or more overlapping shapes.
2. Activate **Shape Builder (S)**. SaVaGe splits them into atomic regions.
3. Regions tint **green** (kept) or **red** (discarded). Click a region to toggle.
4. Press **Enter** (or **Object** → **Commit Shape Builder**, or the sidebar button) to merge kept regions into one path and remove the originals.
5. **Esc** cancels and returns to Select.

Works on closed fills only. Open pencil strokes and live text are not valid sources until you convert them (for text: **Object** → **Convert Text to Outlines**). You can sculpt more than a handful of overlapping shapes in one pass.

5.6 Text — T

- Click empty canvas to place a new text object (starts as “Text”) and type.
- Click an existing text object with the Text tool — or double-click it with Select — to edit it.

Enter or **Esc** finishes editing and keeps what you typed. Click away from the field to do the same.

Typography fields in Props: content, font size, weight. New text uses **DM Sans**.

Convert Text to Outlines: Object menu or the button in Props. SaVaGe turns the letters into real paths using the bundled faces **DM Sans** and **Syne**. Outline late if you still need to edit the live wording. Other font names will outline with the closest bundled face.

6. Panels

6.1 Layers

Hierarchical list of everything on the canvas. Use it for selection, visibility, locking, naming, and reordering. Nested group children appear indented.

6.2 Props (Properties)

Context-sensitive inspector for the **first** selected object.

Paint (fill / stroke)

Type	Behavior
Solid	Single color
Linear	Linear gradient with editable stop colors
Radial	Radial gradient with editable stop colors
Mesh	Grid mesh gradient; edit corner colors
None	No paint

Stroke width uses a slider. Mesh fills look correct on the SaVaGe canvas. Exported SVG uses a native mesh gradient that Inkscape understands. Many web browsers still do not paint mesh fills — keep a **.savage** master or export PNG when the raster look must match.

Path: variable width

For path objects:

- **Taper ends** — applies a width profile along each subpath
- **Clear profile** — removes per-point widths (back to a uniform stroke)

Effects

- **Blur** — softens the object
- **Drop shadow** — offset, blur, color, and opacity

Clip

If the object has a clip mask, release it from Props or **Object** → **Release Clip Mask**. Otherwise a short tip explains how to make one.

6.3 Boards (Artboards)

Multiple artboards live in one document.

Action	How
Activate	Click a board (fits the view to it)
Add	+ or Object → New Artboard
Rename	Double-click the name
Delete	× (at least one board must remain)

The active artboard draws with a stronger accent frame and label. **View** → **Fit All Artboards** frames every board.

6.4 Symbols

Reusable masters stored once; instances on the canvas all follow that master.

Action	How
Create	Select artwork → + in the panel, or Object → Create Symbol
Place	Click a symbol name in the panel
Detach	Select an instance → Detach (panel, Props, or Object → Detach Symbol)
Delete definition	× on the symbol row — placed copies are expanded into ordinary objects first so artwork is not lost

Creating a symbol replaces the selection with an instance of the new master.

6.5 Plug (Plugins)

Lists built-in extras and runs their commands. See [Plugins](#).

6.6 Align & Boolean (sidebar)

Always visible in Edit mode under the right panels.

Align (two or more objects)

Button	Align
L	Left
C	Horizontal center
R	Right
T	Top
M	Vertical middle
B	Bottom

Even spacing (distribute) is not in the sidebar in this release.

Boolean

Button	Operation
Unite	Union
Inter	Intersection
Sub	Subtract (first minus the others)
Xor	Exclusive or / exclude

Live preview: hover a boolean button to see a translucent lime ghost of the result. Click to commit. The preview clears when the pointer leaves the boolean group. The same preview works from **Object** menu items.

Booleans need at least two selected **filled** shapes that can be treated as closed regions (rectangle, ellipse, closed path, or groups of those). Results become path geometry (smooth curves are sampled into polygons). Open strokes are skipped.

7. Menus

7.1 File

Command	Behavior
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Open...	Opens .savage or .svg into Edit. Photos and PNG/JPEG/WEBP/GIF/BMP/TIFF switch to Convert and attach the file. A damaged project file shows an error instead of crashing.
Save Project...	Writes the full document as .savage
Export SVG...	Writes an SVG file
Export PNG...	Writes a PNG snapshot (typically at 2× resolution)

There is no separate “New” command: start from Convert, or **Open...** another file.

7.2 Edit

Command	Behavior
Undo	Ctrl+Z
Redo	Ctrl+Shift+Z / Ctrl+Y
Copy	Ctrl+C
Paste	Ctrl+V

7.3 Object

Command	Behavior
Group / Ungroup	Group the selection; ungroup the primary selection
Unite / Intersect / Subtract / Exclude	Boolean ops + live hover preview
Simplify Path	Reduces complexity of selected paths
Convert Text to Outlines	Turns live text into paths
Make Clip Mask	Last selected shape is the mask; earlier selection is clipped; the mask is hidden
Release Clip Mask	Removes the clip from the selection; the mask reappears only when nothing else still uses it
Create Symbol / Detach Symbol	Symbol workflows
Commit Shape Builder	Finalize a Shape Builder session

New Artboard	Adds a board and fits the view
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Clip mask recipe: select the content first, then the mask shape last → **Make Clip Mask**. The mask must be a path, rectangle, or ellipse. If the command cannot run, a toast explains what to select.

7.4 View

Fit Artboard, Fit All Artboards, Fit Selection, Zoom 50% / 100% / 200%.

7.5 Help

Command	Behavior
User Manual (PDF)...	Opens this guide in your system PDF viewer

8. Advanced editing

8.1 Artboards (multi-board documents)

Treat artboards as design frames (icon set, poster variants, screens). Objects sit in shared drawing space; artboards are frames, not automatic clip parents (clip separately if you need that).

Activate a board before you draw if you want to work in that frame. Save **.savage** when board names, sizes, and layout must survive a reopen. SVG export is the interchange format; PNG is a picture of the current document.

8.2 Symbols

When to use: repeating logos, icons, UI chrome.

- Keep instances linked when many copies should stay identical.
- **Detach** turns one instance into ordinary editable objects.
- Deleting a symbol definition expands remaining instances first.

SVG export writes symbol definitions and instances where possible.

8.3 Shape Builder (professional use)

Ideal for logo marks and icon negative space:

1. Build overlapping primitives.
2. Run Shape Builder; discard outer waste; keep the silhouette regions.
3. Commit once.
4. Optionally Simplify Path and turn remaining pieces into a Symbol.

Very dense stacks may still be faster if you Unite or Subtract first.

8.4 Mesh gradients

1. Select a filled shape.
2. Props → Fill → **Mesh**.
3. Edit corner / control-point color swatches.

The canvas preview matches what you edit. Exported SVG keeps the mesh so Inkscape (and SaVaGe itself) can reopen it. Browsers that do not implement mesh fills will show a missing fill — export PNG when a raster deliverable must match the canvas.

8.5 Variable-width strokes

Uniform strokes use the stroke width in Props. Paths may also store a width at each point.

Method	Result
Props → Taper ends	Automatic profile
Direct Select + Alt-drag	Manual width at a point
Props → Clear profile	Back to uniform

When point widths differ, SaVaGe draws a **filled ribbon** around the centerline. You edit the ribbon by moving points and widths — it is not a separate expanded outline unless you convert it some other way.

8.6 Perspective grid

Toolbar **Persp**:

Mode	Guides
Off	No perspective overlay
1-pt	Horizon + fan from one vanishing point
2-pt	Horizon + fans from two vanishing points

Turn **Snap** on to pull drawing coordinates onto:

- Nearby objects (bounding-box corners and midpoints)
- Path anchors
- Artboard corners and midpoints
- The document grid (when **Grid** is visible) — 32 units
- Perspective rays (when Persp is not Off)

The snap distance stays comfortable at any zoom. Pencil, Brush, Pattern Brush, and Scatter Brush do not snap while you stroke, so freehand stays smooth.

Vanishing points start from the document frame. This release does not include draggable vanishing-point handles.

8.7 Brushes compared

Brush	Geometry	Typical use
Calligraphy (B)	One closed ribbon path	Expressive strokes, ink feel
Pattern	Group of oriented motifs	Borders, decorative paths
Scatter	Group of jittered motifs	Foliage, texture accents

9. Files, export & clipboard

9.1 Formats

Format	Role	What it keeps
.savage	Native project	Everything: artboards, symbols, mesh, effects, widths
.svg	Handoff / web	Solids, linear and radial gradients, and mesh fills that Inkscape can reopen; some browsers skip mesh paint
PNG	Picture export	Visual snapshot
Photos and bitmaps	Convert only	Become vectors after you trace

Opening a **.svg** restores solids, linear/radial gradients, and mesh fills when they are present. Opening a damaged **.savage** file shows an error toast rather than a blank crash.

9.2 Recommended save strategy

1. Always keep a **.savage** working file.
2. Export SVG for handoff to web or other editors.
3. Export PNG for previews, social, or mesh-critical stills.
4. Re-open SVG when you must — colors, gradients, and meshes come back into the document.

9.3 Clipboard

Copy writes:

- SVG text to the system clipboard (when Windows allows it)
- An internal snapshot (including symbol masters used by copied instances)

Paste prefers that internal snapshot, then falls back to SVG on the clipboard. Pasted objects offset slightly so they do not sit exactly on the originals.

9.4 Duplicate

Ctrl+D duplicates the selection in the document (including groups and their children as independent copies). Duplicates sit 16 units down and to the right, next to the original in the same group or layer.

10. Plugins

The **Plug** tab lists extras that ship with SaVaGe.

Plugin	What it does
Duplicate & Offset	Copies selected objects (not groups) 24 units down and right
Randomize Fills	Assigns random palette fills to selected objects that have a fill
Add Guide Rect	Inserts a translucent cyan rectangle as a layout guide

Each command shows a short toast when it finishes. Use **Edit** → **Undo** if you want to reverse it.

11. Keyboard shortcuts cheat sheet

Shortcuts apply when the canvas is focused — not while a text field is active.

Global

Shortcut	Action
V	Select
A	Direct Select
H	Pan
Z	Zoom tool
R	Rectangle
O	Ellipse

L	Line
P	Pen
N	Pencil
B	Calligraphy Brush
S	Shape Builder
T	Text
Space (hold)	Temporary pan
Ctrl+Z	Undo
Ctrl+Shift+Z / Ctrl+Y	Redo
Ctrl+A	Select all
Ctrl+D	Duplicate
Ctrl+C / Ctrl+V	Copy / Paste
Ctrl+0	Fit artboard
Ctrl+1	Zoom 100%
Ctrl+2	Fit selection
Delete / Backspace	Delete selection
Arrows	Nudge 1 px
Shift+Arrows	Nudge 10 px

Polygon, Star, Pattern Brush, and Scatter Brush have no letter shortcut — click them on the tool rail.

Tool-specific

Shortcut	Context	Action
Enter / Esc	Pen	Finish path
Backspace	Pen	Drop last point
Enter	Shape Builder	Commit
Esc	Shape Builder	Cancel
Enter / Esc	Text overlay	Keep the text and close the field

Shift	Select	Add/remove from selection
Shift	Shapes	Constrain (square, circle, 45° line)
Alt	Shape tools	Draw from center
Alt-click	Zoom tool	Zoom out
Alt+drag	Direct Select	Edit point stroke width

Menu items do not show key badges in the title bar; the shortcuts above still work on the canvas.

12. Known limitations & best practices

12.1 Limitations (v0.1)

1. **Distribute / even spacing** is not in the Align bar yet (left/center/right/top/middle/bottom are).
2. Pattern Brush and Scatter Brush spacing and jitter are fixed (no sliders yet).
3. Perspective vanishing points are not draggable.
4. Text outlines use the bundled **DM Sans** and **Syne** faces.
5. Some web browsers will not paint mesh-gradient fills in exported SVG.

12.2 Best practices

Goal	Practice
Logo from PNG	Logo preset → preview → Open in Editor → Simplify → Boolean / Shape Builder → Symbol
Editable master file	Always save .savage
Web SVG	Prefer solids and linear/radial gradients; use PNG for mesh-critical stills
Clean booleans	Work with closed fills; outline text first; skip open pencil strokes
Snappy drawing	Watch the status-bar time; simplify dense converts; hide or delete unused paths
Repeat UI chrome	Symbols over raw copies
Perspective sketching	Persp 1-pt or 2-pt + Snap on while blocking shapes

12.3 Troubleshooting

Symptom	Try
Convert drop does nothing	Drop a PNG, JPEG, WEBP, GIF, BMP, or TIFF — other files are ignored

Converted, but still on Convert	Click Open in Editor (that is expected)
Boolean fails	Select at least two filled closed shapes; ungroup if needed
Shape Builder empty	Select overlapping closed shapes first
Mesh looks empty in a browser	Open it in SaVaGe or Inkscape, or export PNG
Paste did nothing	Click the canvas first; copy from SaVaGe or paste SVG
Open failed	The file may not be a valid .savage project or SVG — check the toast
Heavy lag	Simplify paths; lower convert color precision; hide blur/shadow temporarily
Clicking a tool key types a letter	Click the canvas so a Properties field is not focused

13. Glossary

Term	Meaning in SaVaGe
Artboard	Named frame with size and background; several can live in one file
Boolean	Combine shapes: unite, intersect, subtract, or exclude
Clip mask	One shape hiding parts of others
Instance	A placed copy of a symbol master
Mesh	A grid of colored points interpolated as a fill
Ribbon	A variable-width stroke drawn as a filled outline around a centerline
Shape Builder	Click regions of overlapping shapes to keep or discard them
Speckle	Tiny unwanted islands removed during tracing
.savage	SaVaGe's native project file (full document)

Appendix A — Suggested learning path

1. Convert a simple logo PNG; use **Open in Editor**.
2. Recolor with Props; rename layers.
3. Draw primitives; Unite; Shape Builder.
4. Create a Symbol; place twice; detach one.
5. Try Brush, Pattern, and Scatter.

6. Enable 1-pt perspective and sketch with Snap.
 7. Save **.savage** ; export SVG and PNG; reopen both and compare.
 8. Run Plug → Randomize Fills once for fun, then Undo.
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Appendix B — Product identity

SaVaGe blends **SVG** with a bold craft aesthetic: graphite interface, lime accent, and expressive typography (Syne / DM Sans).

For help in the app, use **Help** → **User Manual (PDF)**....

End of user manual.